

Deployment Guide

Serverless Deployment to Vercel

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Vercel Serverless Deployment Guide

This guide details the prerequisites, local verification steps, and deployment workflow for serving the CreditGuard AI Flask application on Vercel's serverless platform.

Deployment Prerequisites

- Vercel CLI installed (`npm install -g vercel`)
 - Python 3.10+ environment installed locally
 - Git tracking main branch
-

Configuration Files

1. Vercel Configuration (`vercel.json`)

The application is configured to build using Vercel's Python builder.

```
{
  "version": 2,
  "builds": [
    {
      "src": "index.py",
      "use": "@vercel/python",
      "config": {
        "excludeFiles":
          "{data,notebooks,logs,diagrams,tests,docs,demo,interview,reports,resume,screenshots,src,credit_card_approval_prediction.egg-info,.pytest_cache,__pycache__,.github,.git,models/artifacts,models/trained}/"
      }
    }
  ],
  "routes": [
    {
      "src": "/static/(.*)",
      "dest": "index.py"
    },
    {
      "src": "/(.*)",
      "dest": "index.py"
    }
  ]
}
```

2. Ignore Configurations (`vercelignore`)

To bypass Vercel's standard 250 MB size limit, non-essential files are excluded during upload:

```
data/  
notebooks/  
logs/  
diagrams/  
tests/  
docs/  
/app.py
```

Local Verification Workflow

Before deploying, always test the serverless configuration locally:

1. Verify relative imports:

```
python -c "import sys; sys.path.insert(0, '.'); from app.app import app; print('Imports pass')"
```

2. Launch the server locally:

```
python index.py
```

3. Check that the console prints model pre-loading success statements and serves on `http://127.0.0.1:5000`.

Deployment Commands

Deploy the optimized application using Vercel CLI:

```
# Deploy preview build  
vercel  
# Promote to production  
vercel --prod
```

Once complete, the CLI will output the URL and verify that the page returns `HTTP 200`.